4 **C** Since drone hovers steadily and remains level, the thrust from each propeller should be equal.

$$\text{Thrust from each propeller} = \frac{1.20 \times 9.81}{4} = 2.943 \text{ N}$$

$$F = \frac{dp}{dt} = \frac{m}{t}(v - 0)$$

$$2.943 = (0.50)v$$

$$v = 5.886 = 5.9 \text{ m s}^{-1}$$

5 **A** Since the resultant force on the ball is zero, it falls with constant velocity. Thus, its kinetic